

ILO: Understand TCP packets transfer through terminal command line in LINUX/POSIX TERMINAL SYSTEMS

Supplementary Handout summarizing the useful commands from the Red Hat tutorial

IT 213: Supplementary Handout - Tcpdump Cheat Sheet

Topic: Advanced Packet Filtering & Logic

Source: Red Hat Administration Guide [<https://www.redhat.com/en/blog/filtering-tcpdump>]

Student Toolkit

Keep this page! These commands are standard for Linux System Administrators.

1. The Basics (Filtering by Target)

Instead of capturing *everything* (which is messy), tell **tcpdump** exactly who you want to listen to.

Goal	Command	Explanation
Filter by IP	<code>tcpdump host 1.2.3.4</code>	Only show packets coming from OR going to this IP.
Filter by Name	<code>tcpdump host google.com</code>	Automatically finds the IP for Google and filters for it.
Filter by Network	<code>tcpdump net 192.168.1.0/24</code>	Watch an entire subnet (e.g., the whole Lab room).
Filter by Port	<code>tcpdump port 80</code>	Only show Web traffic (HTTP). Ignore everything else.

2. Directional Filtering (Source vs. Destination)

Sometimes you only care about what **YOU** sent, or what the **SERVER** sent back.

Goal	Command	Explanation
Outgoing Only	<code>tcpdump src 192.168.1.5</code>	"Show me only what I sent." (Source = Me)
Incoming Only	<code>tcpdump dst 192.168.1.5</code>	"Show me only what I <i>received</i> ." (Destination = Me)

3. Logical Operators (The Power Moves)

You can combine rules using and, or, and not (or !).

- **Tip:** If using () or !, put the whole query in quotes '...' so the terminal doesn't get confused.

Goal	Command	Logic
Specific Conversation	<code>tcpdump host 8.8.8.8 and port 53</code>	"Must be Google DNS AND must be DNS traffic."
Exclude SSH	<code>tcpdump not port 22</code>	"Show me everything EXCEPT my admin connection."
Complex Filter	<code>tcpdump 'src 10.0.0.1 and (port 80 or port 443)'</code>	"Show me web traffic (HTTP or HTTPS) coming from this specific IP."

4. Saving Data (Pcap vs Text)

In Lab 5, we used > to save **Text**. Real pros use -w to save **Raw Packets** (PCAP) that can be opened in Wireshark later.

- **Save to Wireshark Format:**

`sudo tcpdump -w my_capture.pcap port 80`

(Nothing prints to screen. It saves directly to the file).

- **Read it back later:**

`tcpdump -r my_capture.pcap`

Why these specific commands?

1. **host vs net:**

- **Pitfall:** You should not ping a whole room (**broadcast**). `tcpdump net` is the diagnostic tool for: "**Is anyone in the lab transmitting?**"

2. **port Filtering:**

This is crucial when working on remote servers (SSH). **If you run tcpdump without not port 22, the output will loop infinitely because tcpdump captures its own text** stream being sent over SSH!

- *Rule of Thumb:* **Always tcpdump not port 22** if you are logged into a cloud server.

3. **Logical Operators (and/or):**

- **Syntax Note:** The single quotes '...' are critical when using parentheses or the bang ! symbol in Linux terminals. Without quotes, bash interprets ! as "history expansion" and crashes the command.

4. **-w (Write) vs > (Redirect):**

- **> (Text):** Good for grep, bad for Wireshark. It only saves what you saw on screen.
- **-w (PCAP):** Good for Wireshark. It saves the *entire* packet binary (headers + payloads). This allows you to open the file in Wireshark later and drill down into details that text doesn't show.